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Persistent and Transitory Components of Firm Characteristics: Implications for Asset Pricing

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1 Big picture

- **Question.** How should we interpret cross-sectional return predictability when firm characteristics contain both persistent and transitory components?
- **Central debate.** A recent literature argues that return predictability is mostly driven by transitory characteristic components. This paper argues that the compensation for persistent and transitory components varies across characteristics, both in magnitude and sign.
- **Key decomposition.** Each firm characteristic is written as a sum of a persistent component and a transitory component:
$$X_t = X_t^P + X_t^T.$$
- **Expected-return model.** Expected returns compensate market beta plus the persistent and transitory components:
$$E_t(R_{t+1}) = \lambda_M \beta_M + \lambda_P X_t^P + \lambda_T X_t^T.$$
- **Horizon insight.** If expected returns depend on the same characteristic that defines the portfolio, returns after formation should decay at the same rate as the characteristic and its factor exposure. If returns decay faster or slower than this benchmark, the persistent and transitory components are compensated differently.
- **Main test.** The authors compare returns of old characteristic sorts to the newest sort. Old sorts are portfolios formed using characteristics measured 12, 24, 36, 48, or 60 months earlier, while returns are measured today.
- **Core empirical finding.** Old-versus-new alphas vary widely across characteristics. Some old sorts earn positive alphas relative to the newest sort, while others earn negative alphas.
- **Economic interpretation.** Positive old-versus-new alphas imply that the persistent component is relatively more compensated. Negative alphas imply that the transitory component is relatively more compensated.
- **Trading implication.** Combining newest sorts with old sorts often improves Sharpe ratios substantially. The old information contains economically useful information about the source of a characteristic signal.
- **Old-minus-new stocks.** Within today's extreme characteristic portfolios, the paper separates stocks that were already in similar extreme portfolios years earlier from stocks that are newly extreme today. This yields another decomposition of persistent and transitory characteristic information.
- **Factor-model implication.** Benchmark factor models that price newest sorts often fail to price old-versus-new strategies. Thus, a model that explains short-horizon characteristic returns may still fail at longer horizons.
- **Discount-rate implication.** Characteristics with strong persistent compensation imply larger long-term discount-rate differences across firms. This matters for firm valuation and for interpreting characteristics as risk or mispricing.
- **Takeaway.** Characteristic-based predictability is not one-dimensional. The same current characteristic value can mean different expected returns depending on whether it comes from a persistent or transitory component.

2 Data and measurement

2.1 Characteristics and sample

- The paper studies 56 firm characteristics widely used in empirical asset pricing.

- The sample period runs from July 1972 through December 2019.
- The authors construct characteristic-sorted long-short decile portfolios.
- For characteristics where the sign of predictability is negative, the characteristic is multiplied by -1 so that higher values correspond to higher expected returns.
- The main illustrative characteristics are size, book-to-market, profitability, and investment.

Interpretation: The sign normalization allows all characteristics to be interpreted in the same direction: the high-minus-low portfolio is expected to earn a positive return when the characteristic is predictive.

2.2 Newest and old sorts

For each characteristic X, the paper forms high-minus-low decile portfolio returns:

$$R_{X,(t-s),t+1} = R^{High}_{X,(t-s),t+1} - R^{Low}_{X,(t-s),t+1}. \tag{1}$$

- The first subscript X denotes the characteristic.
- The second subscript $(t-s)$ denotes the sorting date.
- The third subscript $t+1$ denotes the return observation date.
- $s = 0$ is the newest sort.
- $s = 12, 24, 36, 48, 60$ months are old sorts.

Interpretation: An old sort tracks whether a characteristic measured in the past continues to predict returns today. It therefore reveals whether predictability is persistent or transitory.

2.3 Characteristic-matched returns

For the motivating figures, the paper compares actual portfolio returns to characteristic-matched returns. The characteristic-matched return tracks what would happen if returns decayed exactly like the characteristic ranks themselves.

Interpretation: If actual returns decay at the same speed as characteristic-matched returns, old sorts behave like a standard characteristic-based expected-return model. If actual returns decay faster or slower, then persistent and transitory components are compensated differently.

2.4 Old-versus-new alphas

The unconditional old-versus-new alpha is estimated from:

$$R_{X,(t-s),t+1} = \alpha^u_s + \beta^u_s R_{X,(t),t+1} + \varepsilon_{X,(t-s),t+1}. \tag{2}$$

The conditional alpha adjusts for the decay in exposure using rolling regressions:

$$\alpha^c_s = E\left(R_{X,(t-s),t+1}\right) - \beta^c_s E\left(R_{X,(t),t+1}\right), \tag{3}$$

where β^c_s is estimated from rolling regressions of the old sort on the newest sort.

Interpretation: The beta controls for the fact that old sorts usually have lower exposure to the characteristic factor than newest sorts. The alpha asks whether the return of the old sort is high or low relative to its remaining exposure.

2.5 Old-minus-new stocks

Within today's high and low characteristic portfolios, stocks are split into old and new groups.

$$R^{New}_{X,(t),t+1} = R^{High,New}_{X,(t),t+1} - R^{Low,New}_{X,(t),t+1}, \tag{4}$$

$$R^{Old}_{X,(t),t+1} = R^{High,Old}_{X,(t),t+1} - R^{Low,Old}_{X,(t),t+1}. \tag{5}$$

- New stocks are currently in an extreme characteristic portfolio but were not extreme in the same direction in the past.
- Old stocks are currently in an extreme characteristic portfolio and were also extreme in the same direction in the past.
- The old-minus-new return compares persistent characteristic signals with newly extreme signals.

Interpretation: This decomposition is a direct trading comparison: among stocks with the same current characteristic spread, are returns higher for firms with persistent or newly emerging characteristic values?

2.6 Principal components

The paper extracts principal components from the old and newest sorts. A representative regression is:

$$\gamma'_{(t-s),z} R_{X,(t-s),t+1} = \alpha_{s,z} + \beta'_{s,z} R_{PC3,(t),t+1} + \varepsilon_{(t-s),z,t+1}, \tag{6}$$

where $z = 1, 2, 3$ and $R_{PC3,(t),t+1}$ contains the first three principal components of newest sorts.

Interpretation: Principal components aggregate many characteristic portfolios and test whether a statistical factor model extracted from newest sorts can price old sorts.

2.7 Long-term discount rates

The paper follows a Gordon-growth logic:

$$P = \frac{D}{r - g}, \tag{7}$$

where P is price, D is the implied cash-flow level, g is growth, and r is the implied discount rate.

Interpretation: If characteristics predict returns persistently, then discount-rate differences across firms should be long-lived and economically meaningful.

3 Model and empirical predictions

3.1 Persistent-transitory decomposition

The model writes a characteristic as:

$$X_t = X^P_t + X^T_t, \tag{8}$$

where X^P_t is persistent and X^T_t is transitory.

The expected return is:

$$E_t(R_{t+1}) = \lambda_M \beta_M + \lambda_P X^P_t + \lambda_T X^T_t. \tag{9}$$

Interpretation: The key parameters are λ_P and λ_T. If they are equal, the origin of a characteristic value does not matter. If they differ, current characteristic values with different histories imply different expected returns.

3.2 Null hypothesis: standard characteristic-based model

The null is:

$$\lambda_P = \lambda_T > 0. \tag{10}$$

- Both persistent and transitory components are compensated equally.
- Expected returns decay at the same speed as factor exposures.
- Old-versus-new alphas should be approximately zero.

Interpretation: Under this null, the newest sort and old sorts are just different exposures to the same priced factor. A factor model using newest sorts should price old sorts.

3.3 Alternative 1: only the transitory component is compensated

$$\lambda_P = 0, \qquad \lambda_T > 0. \tag{11}$$

- Expected returns decay faster than characteristic exposure.
- Old sorts become less profitable relative to their remaining factor exposure.
- Old-versus-new alphas should be negative.

Interpretation: This is the view that characteristics mainly capture short-lived mispricing or transitory return predictability.

3.4 Alternative 2: only the persistent component is compensated

$$\lambda_P > 0, \qquad \lambda_T = 0. \tag{12}$$

- Expected returns decay more slowly than characteristic exposure.
- Old sorts retain profitability beyond what their exposure to the newest sort would imply.
- Old-versus-new alphas should be positive.

Interpretation: This is the view that persistent characteristic components reflect more durable risk or valuation differences.

3.5 Analytical prediction for the newest sort

The expected return of the newest sort is approximately:

$$E(R_{X,(t),t+1}) = 2 \times 1.755 \times \left(Var(X_i^P)\lambda_P + \left[1 - Var(X_i^P)\right]\lambda_T\right). \tag{13}$$

Interpretation: The newest sort combines persistent and transitory components. Its expected return is a mixture of compensation for both components.

3.6 Analytical prediction for old sorts

The expected return of the old sort is approximately:

$$E(R_{X,(t-s),t+1}) = 2 \times 1.755 \times Var(X_i^P)\rho^s\lambda_P. \tag{14}$$

Interpretation: The old sort mostly preserves the persistent component because the transitory component decays quickly. Therefore old sorts are informative about λ_P.

3.7 Alpha prediction

The model implies that the old-versus-new alpha is related to:

$$\alpha_s^u \propto \rho^s(\lambda_P - \lambda_T). \tag{15}$$

- If $\lambda_P = \lambda_T$, alphas should be near zero.
- If $\lambda_P < \lambda_T$, alphas should be negative.
- If $\lambda_P > \lambda_T$, alphas should be positive.

Interpretation: The sign of the old-versus-new alpha directly identifies whether the persistent or transitory component is more compensated.

4 Identification and empirical design

4.1 What identifies the main result

- The empirical variation comes from comparing returns measured today for portfolios sorted at different dates in the past.
- The newest sort controls for the current characteristic signal.
- Old sorts reveal whether historical characteristic values continue to predict returns after current exposure is controlled for.
- Cross-characteristic variation shows whether persistent or transitory components dominate for different characteristics.

Interpretation: The design is not based on an external shock. It is a horizon-based test of whether characteristic predictability decays like factor exposure.

4.2 Why alphas rather than raw average returns

- Characteristics vary widely in persistence.
- Simply comparing average returns of newest and old sorts would confound compensation with exposure decay.
- Regressing old sorts on newest sorts controls for remaining exposure to the characteristic factor.
- The residual alpha captures excess performance relative to the characteristic exposure.

Interpretation: The alpha is the clean object because it adjusts for the fact that an old book-to-market sort and an old profitability sort may retain very different amounts of their original characteristic exposure.

4.3 Why old-minus-new stocks matter

- New and old stocks have the same current characteristic portfolio membership.
- They differ in whether the characteristic signal is newly extreme or persistent.
- Comparing these groups tests whether characteristic history matters conditional on current characteristic value.

Interpretation: This test translates the abstract persistent-transitory decomposition into a portfolio construction rule that can be implemented directly.

4.4 Benchmark factor-model tests

- The paper tests whether standard factor models price old-versus-new sorts and old-minus-new strategies.
- Models include the CAPM, Fama-French three-factor, Fama-French five-factor, and Fama-French five-factor plus momentum specifications.
- If newest sorts are valid characteristic factors, they should price old sorts under the null.
- The paper finds large alphas, especially for components whose returns decay too slowly or too quickly relative to benchmark exposures.

Interpretation: The failure of benchmark models means that explaining short-horizon characteristic returns is not enough. A successful model must price returns across horizons.

4.5 Limitations

- The decomposition into persistent and transitory components is model-based and implemented through horizon dynamics, not by observing the true components directly.
- Characteristic definitions and signs are standardized, but some characteristics may contain multiple economic channels.
- Factor-model failures can reflect both omitted risk and mispricing.
- The interpretation of long-term discount rates depends on assumptions about growth and the mapping from returns to price levels.

Interpretation: The paper's strength is the disciplined horizon test. Its limits come from the broader ambiguity of whether characteristic returns reflect risk, mispricing, or both.

5 Tables and figures

5.1 Table 1 and Figure 1: motivating evidence from popular characteristics

Read:

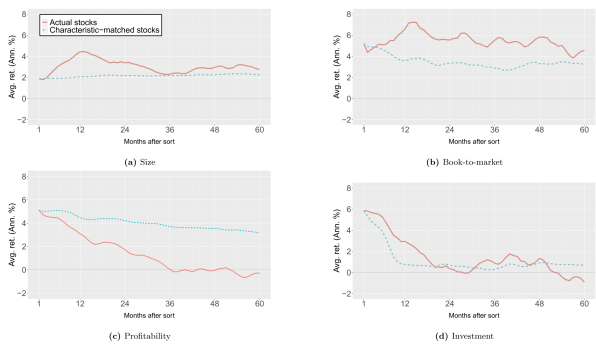
- Table 1 summarizes old and new sorts for size, book-to-market, profitability, and investment.
- Panel A shows average returns after formation. Book-to-market returns are high and persistent, while profitability and investment returns decay faster.
- Panel B shows old-versus-new alphas. Book-to-market has positive alphas at multiple horizons, while profitability and investment display negative alphas at long horizons.
- Panel C shows that combining old and newest sorts can improve Sharpe ratios, especially for book-to-market.
- Figure 1 plots actual returns and characteristic-matched returns, showing whether returns decay faster or slower than the characteristic itself.

Economic message: The core empirical fact is already visible in the four benchmark characteristics: some return predictability is persistent, while other return predictability is transitory.

Table 1

Old and new sorts on popular characteristics. This table reports summary statistics for old and new sorts on size, book-to-market, profitability, and investment. We track the returns of long-short decile portfolios (value-weighted and split at NYSE breakpoints) for each characteristic from one month to five years after portfolio formation. Panel A shows the average high-minus-low return. Panel B reports the alpha of old with respect to new sorts. The unconditional alpha, α^u, is the intercept from a regression of the return of an old sort on the contemporaneous return of the newest sort: $R_{X,(t-s),t+1} = \alpha_s^u + \beta_s^u R_{X,(t),t+1} + \epsilon_{X,(t-s),t+1}$ (see Eq. (2)). The conditional alpha, α^c, is calculated as the average return of a strategy that invests in $R_{X,(t-s),t+1}$ and hedges in each month t the conditional exposure to $R_{X,(t),t+1}$. Following Eq. (3), we estimate this exposure over a 36 month historical rolling window. The t-statistics are calculated using White (1980) heteroskedasticity-consistent standard errors. Panel C reports the Sharpe ratio of the newest sort, Sharpe($R_{X,(t),t+1}$), and the maximum increase in Sharpe ratio achievable from combining the newest sort with the older sorts (based on either the unconditional or conditional alpha). The sample period runs from July 1972 through December 2019.

Horizon s	Size		Book-to-market		Profitability		Investment	
	Panel A: Average return of new and old sorts ($R_{X,(t-s),t+1}$)							
	Avg.	t-stat	Avg.	t-stat	Avg.	t-stat	Avg.	(t-stat)
0	2.01	(0.86)	5.24	(1.61)	5.09	(3.32)	6.06	(3.62)
12	4.45	(1.91)	7.15	(3.14)	2.94	(2.05)	2.86	(1.99)
24	3.48	(1.50)	5.49	(2.62)	1.50	(1.05)	0.47	(0.31)
36	2.44	(1.13)	5.54	(2.75)	-0.02	(-0.01)	0.83	(0.53)
48	3.08	(1.48)	5.97	(3.03)	0.16	(0.11)	1.02	(0.62)
60	2.54	(1.17)	4.44	(2.23)	-0.35	(-0.23)	-0.78	(-0.47)
	Panel B: Alphas of old-versus-new sorts							
	α_s^u	β_s^u	α_s^c	α_s^u	β_s^u	α_s^c	α_s^u	β_s^u	α_s^c	α_s^u	β_s^u	α_s^c
12	2.63	0.90	3.27	4.66	0.48	4.78	-1.09	0.79	-1.02	0.49	0.39	0.65
	(2.74)	(20.78)	(3.03)	(2.95)	(8.31)	(2.79)	(-1.42)	(15.73)	(-1.42)	(0.38)	(10.82)	(0.49)
24	1.77	0.85	2.65	3.66	0.35	4.17	-1.84	0.66	-1.67	-1.15	0.27	-1.43
	(1.53)	(18.57)	(2.14)	(2.14)	(7.07)	(2.23)	(-1.83)	(10.67)	(-1.75)	(-0.79)	(6.06)	(-0.98)
36	0.83	0.80	1.71	4.15	0.26	4.40	-3.11	0.61	-3.25	-0.79	0.27	-1.59
	(0.78)	(21.11)	(1.54)	(2.35)	(5.31)	(2.28)	(-2.72)	(11.02)	(-3.02)	(-0.52)	(6.70)	(-1.04)
48	1.56	0.76	2.35	4.65	0.25	4.64	-2.81	0.58	-2.83	-1.30	0.38	-1.77
	(1.43)	(20.03)	(2.10)	(2.66)	(5.43)	(2.41)	(-2.46)	(10.95)	(-2.69)	(-0.85)	(7.45)	(-1.16)
60	1.03	0.75	1.84	3.18	0.24	2.71	-3.18	0.56	-3.16	-2.85	0.34	-3.53
	(0.83)	(15.84)	(1.40)	(1.78)	(5.59)	(1.37)	(-2.64)	(8.48)	(-2.81)	(-1.79)	(5.65)	(-2.30)
	Panel C: Improvements in Sharpe ratio							
	Sharpe ratio of the newest sort ($R_{X,(t),t+1}$)							
0	0.12		0.23		0.48		0.52	
	Max. Sharpe($R_{X,(t-s),t+1}$, $R_{X,(t),t+1}$) - Sharpe($R_{X,(t),t+1}$)							
	u	c	u	c	u	c	u	c
12	0.28	0.34	0.23	0.25	0.04	0.03	0.00	0.00
24	0.12	0.22	0.15	0.21	0.07	0.04	0.01	0.01
36	0.04	0.14	0.17	0.22	0.14	0.13	0.01	0.01
48	0.11	0.21	0.21	0.22	0.12	0.11	0.01	0.02
60	0.05	0.12	0.11	0.10	0.13	0.12	0.06	0.09



(a) Size (b) Book-to-market (c) Profitability (d) Investment

Fig. 1. Returns and characteristic-matched returns We plot annualized average returns and characteristic-matched average returns of the high-minus-low decile portfolios sorted on size, book-to-market, profitability, and investment. We consider horizons from one month up to five years after portfolio formation. The sample period runs from July 1972 through December 2019.

5.2 Table 2: model simulations

Read:

- Table 2 compares simulated alpha distributions under the null and the two alternatives.
- Under the null, the alpha distribution is centered near zero.
- If only the transitory component is compensated, old-versus-new alphas are negative.
- If only the persistent component is compensated, old-versus-new alphas are positive.
- The empirical distribution has both large negative and large positive alphas, inconsistent with one uniform compensation pattern.

Economic message: The model gives a clear interpretation of the data: different characteristics reward persistent and transitory components differently.

Table 2

Old-versus-new sorts and stocks in model simulations. This table reports results from 10,000 simulations of the model presented in Section 4. In Panel A we present the 5th, 50th, and 95th percentiles of the simulated distribution of the alpha from a regression of a three-year old sort on the newest sort (see Eq (15)). In Panel B, we present the same distribution for the average return of old-minus-new stocks (as defined in Section 4.6). We also report in each panel (i) the analytical estimate, derived from a setting with IID idiosyncratic return shocks and (ii) the distribution of the estimate in the data, derived from the large set of 56 characteristics. For the model simulations, we consider three cases. Under the null of a standard characteristic-based model of expected returns, the compensation for the transitory and persistent component of the characteristic is equal: $\lambda_P = \lambda_T = 3.17\%$. Under the first alternative, only the transitory component is compensated in expected returns: $\lambda_P = 0,\ \lambda_T = -4.49\%$. Under the second alternative, only the persistent component is compensated in expected returns: $\lambda_P = 4.49\%,\ \lambda_T = 0$.

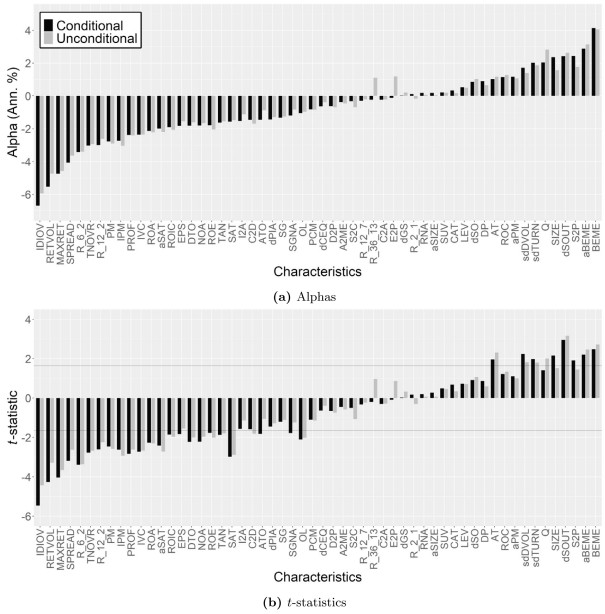
		5th	50th	95th	Analytical
		Panel A: Alpha of old-versus-new sort			
Null:	$\lambda_P = \lambda_T > 0$	-2.46%	0.02%	2.53%	0.00%
Only transitory:	$\lambda_T > \lambda_P = 0$	-4.73%	-2.25%	0.31%	-2.42%
Only persistent:	$\lambda_P > \lambda_T = 0$	-0.27%	2.24%	4.79%	2.42%
Empirical		-4.67%	-0.76%	3.02%	
		Panel B: Average return of old-minus-new stocks			
Null:	$\lambda_P = \lambda_T > 0$	-1.58%	0.52%	2.57%	0.54%
Only transitory:	$\lambda_T > \lambda_P = 0$	-3.89%	-1.79%	0.34%	-1.89%
Only persistent:	$\lambda_P > \lambda_T = 0$	0.42%	2.50%	4.57%	2.65%
Empirical		-4.86%	-0.22%	5.90%	

5.3 Figure 2 and Figure 3: old-versus-new alphas and Sharpe improvements

Read:

- Figure 2 ranks characteristics by conditional old-versus-new alpha.
- Many characteristics have large negative alphas, and many have positive alphas.
- Figure 3 shows that combining old sorts with newest sorts can produce substantial Sharpe-ratio improvements.
- The improvements are often economically large for characteristics with strongly nonzero alphas.

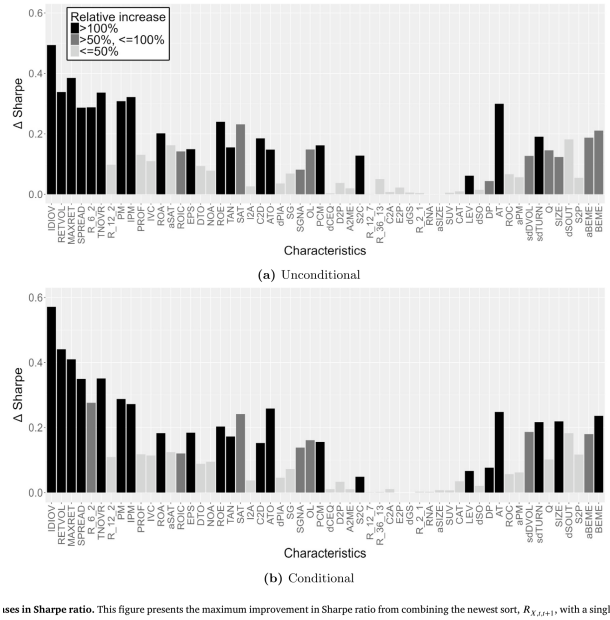
Economic message: Horizon information is not just statistically interesting. It can improve portfolio construction by revealing whether old or new characteristic information is more valuable.



(a) Alphas

(b) t-statistics

Figure 2: alphas of old-versus-new sorts



(a) Unconditional

(b) Conditional

ases in Sharpe ratio. This figure presents the maximum improvement in Sharpe ratio from combining the newest sort, $R_{X,L+1}$, with a single

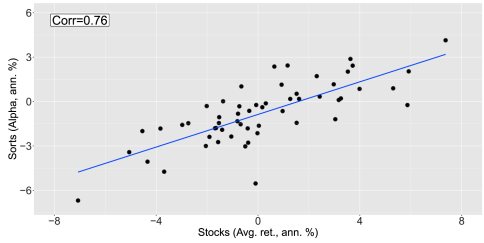
Figure 3: increases in Sharpe ratio

5.4 Figure 4: old-minus-new stocks and PC1 loadings

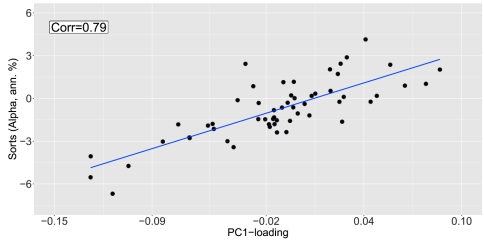
Read:

- Panel A compares conditional old-versus-new alphas with average returns of old-minus-new stock strategies.
- The correlation is 0.76, showing that the stock-level decomposition aligns with the portfolio-level alpha evidence.
- Panel B compares conditional alphas with PC1 loadings and finds a correlation of 0.79.
- Characteristics whose old sorts load strongly on the first principal component tend to have stronger old-versus-new performance.

Economic message: The old-minus-new stock comparison validates the persistent-transitory interpretation at the stock level, not only at the portfolio-sort level.



(a) Old-versus-new sorts and old-minus-new stocks



(b) Old-versus-new sorts and loadings of PC1

5.5 Table 3 and Table 4: principal components and benchmark factor models

Read:

- Table 3 shows that principal components extracted from newest sorts do not price principal components extracted from old sorts.
- PC1 of old sorts earns large positive alphas, especially at long horizons.
- Table 4 shows that benchmark factor models struggle to price old-versus-new PC strategies.
- The CAPM, Fama-French three-factor, five-factor, and five-factor plus momentum models leave large alphas for old sorts.

Economic message: Factor models built from short-horizon characteristic returns fail to price horizon dynamics. A model must price old and new characteristic information jointly.

Table 3

Principal components of new and old sorts. Panel A of this table reports the intercept (α) and associated t-statistic (based on White (1980) heteroskedasticity-consistent standard errors) from regressing the first three principal components of old sorts ($\gamma'_{(t-s),z} R_{X,(t-s),t+1}$, $z = 1, 2, 3$ and $s > 0$) on a statistical factor model containing the first three principal components of new sorts ($R_{PC3,(t),t+1} = [\gamma_{(t),1}, \gamma_{(t),2}, \gamma_{(t),3}]' R_{X,(t),t+1}$). The last column of this panel reports the associated F-statistics and p-values from a (conditional heteroskedasticity-consistent) GRS test. Panel B is identical to Panel A except that we now apply the loadings of the principal components of the newest sorts to each of the old sorts. Hence, the test asset returns are defined as: $\gamma'_{(t),z} R_{X,(t-s),t+1}$. Panel C reports summary statistics for the first principal component of new and old sorts reported in Panel A. The sample period runs from July 1972 through December 2019.

	PC1 ($z = 1$)		PC2 ($z = 2$)		PC3 ($z = 3$)		GRS test	
Horizon s	α	t-stat	α	t-stat	α	t-stat	F-stat	p-val
	Panel A: PCs of old sorts on statistical factor model ($\gamma'_{(t-s),z} R_{X,(t-s),t+1}$ on $R_{PC3,(t),t+1}$)							
12	3.11	(3.61)	0.22	(0.31)	1.29	(0.95)	4.42	0.00
24	5.03	(4.89)	0.44	(0.40)	0.36	(0.27)	8.41	0.00
36	5.27	(5.08)	-0.58	(-0.46)	0.62	(0.45)	8.92	0.00
48	5.72	(5.12)	-1.31	(-1.02)	0.95	(0.71)	9.87	0.00
60	5.45	(4.84)	-1.08	(-0.79)	0.14	(0.11)	8.92	0.00
	Panel B: PCs of old sorts with loadings fixed ($\gamma'_{(t),z} R_{X,(t-s),t+1}$ on $R_{PC3,(t),t+1}$)							
12	2.89	(3.49)	0.17	(0.25)	-2.43	(-2.00)	5.50	0.00
24	4.45	(4.41)	1.51	(1.54)	-1.35	(-1.10)	8.05	0.00
36	4.84	(4.41)	1.57	(1.30)	0.18	(0.16)	8.12	0.00
48	5.75	(5.18)	0.79	(0.65)	-1.19	(-1.10)	10.22	0.00
60	5.36	(4.72)	1.31	(1.02)	-1.93	(-1.64)	9.41	0.00

Panel C: Summary statistics for first principal components ($\gamma'_{(t-s),z} R_{X,(t-s),t+1}$)

	Avg. Ret.	Correlations				
		0	12	24	36	48
0	-3.76					
12	0.11	0.91				
24	1.90	0.86	0.98			
36	1.94	0.81	0.95	0.98		
48	2.10	0.78	0.92	0.95	0.98	
60	1.70	0.75	0.90	0.93	0.96	0.98

Table 3: principal components of new and old sorts

new sorts in benchmark factor models. This table presents average returns as well as the intercept (α), the associated t-statis
n White (1980) heteroskedasticity-consistent standard errors), and the factor contribution ($\beta'_{s,1}\mu_F$) from regressing the first princi
nt (PC1) of new and old sorts on benchmark factor models. In Panel A, we consider each horizon in isolation and define the retu
st asset as $\gamma'_{(t),1} R_{X,(t-s),t+1}$ for $s \geq 0$. In Panel B, we look at old-minus-new strategies where we define the return of the test asset
$_{(t-s),t+1} - R_{X,(t),t+1})$ for $s > 0$. The sample period runs from July 1972 through December 2019.

		CAPM			FF3M			FF5M			FF5M+MOM		
s	Avg. ret.	t-stat	α	t-stat	$\beta'_{s,1}\mu_F$	α	t-stat	$\beta'_{s,1}\mu_F$	α	t-stat	$\beta'_{s,1}\mu_F$	α	t-stat	$\beta'_{s,1}\mu$
	Panel A: PC1 of new and old sorts ($\gamma'_{(t),1} R_{X,(t-s),t+1}$)													
	-3.76	(-1.06)	-10.15	(-3.55)	6.39	-9.43	(-4.82)	5.67	-2.72	(-1.68)	-1.03	0.20	(0.13)	-3.96
	1.03	(0.33)	-4.32	(-1.67)	5.35	-3.33	(-2.05)	4.36	1.78	(1.33)	-0.75	2.14	(1.59)	-1.11
	2.84	(0.96)	-1.95	(-0.78)	4.79	-0.55	(-0.35)	3.39	3.90	(2.81)	-1.06	3.86	(2.72)	-1.01
	2.98	(1.07)	-1.54	(-0.64)	4.53	0.14	(0.09)	2.85	4.64	(3.45)	-1.65	4.54	(3.27)	-1.55
	3.87	(1.44)	-0.54	(-0.23)	4.41	1.16	(0.80)	2.70	5.59	(4.33)	-1.72	5.58	(4.22)	-1.71
	3.46	(1.34)	-0.80	(-0.36)	4.26	0.66	(0.46)	2.80	4.83	(3.75)	-1.36	4.93	(3.65)	-1.47
	Panel B: PC1 of old-minus-new sorts ($\gamma'_{(t),1}(R_{X,(t-s),t+1} - R_{X,(t),t+1})$)													
	4.79	(4.11)	5.83	(5.20)	-1.05	6.10	(5.67)	-1.31	4.50	(3.83)	0.28	1.93	(1.85)	2.85
	6.60	(4.51)	8.20	(6.15)	-1.60	8.88	(6.91)	-2.28	6.62	(4.67)	-0.02	3.65	(2.92)	2.95
	6.74	(4.11)	8.60	(5.74)	-1.86	9.57	(6.60)	-2.83	7.36	(4.80)	-0.62	4.33	(3.13)	2.41
	7.62	(4.50)	9.61	(6.28)	-1.99	10.59	(7.35)	-2.97	8.31	(5.45)	-0.69	5.38	(3.73)	2.25
	7.22	(4.10)	9.35	(5.96)	-2.13	10.09	(6.85)	-2.87	7.55	(5.04)	-0.33	4.73	(3.24)	2.49

Table 4: old and new sorts in benchmark factor models

5.6 Table 5 and Table 6: fast versus slow decay

Read:

- Table 5 decomposes characteristics into groups where returns decay too fast versus too slow.
- The fast-decay group earns negative alphas; the slow-decay group earns positive alphas.
- Table 6 uses old and new stock strategies in benchmark factor models.
- The old-minus-new stock strategy earns positive average returns and significant alphas, especially when aggregated using PC1 loadings.

Economic message: The results are not driven by a few individual characteristics. Aggregated fast- and slow-decay strategies preserve the same economic pattern.

Table 5

Old-versus-new sorts when returns decay fast versus slow. This table presents average returns and the intercept (α) with associated t-statistic (based on White (1980) heteroskedasticity-consistent standard errors) from standard factor regressions. In Panel A, the test asset is based on the return of the old-versus-new sort strategy that is the focus in our model and aggregates across characteristics using the PC1 loadings: $\gamma'_{(t),1}(R_{X,(t-36),t+1} - \beta_{36} R_{X,(t),t+1})$. We focus on the three-year horizon and present results for both an unconditional and conditional specification of β_{36} (β^u_{36} from Eq. (2) and $\beta^c_{36,t}$ from Eq. (3)). In Panel B and C, we decompose the test asset returns into the part coming from the characteristics with returns that decay too fast versus too slow, which are those characteristics on which PC1 loads with a negative versus positive sign (see Eqs. (19) and (20)). Because the absolute sum of negative loadings is about twice as large as of positive loadings, we rescale the latter such that the two test asset returns have the same volatility. The sample period runs from July 1972 through December 2019.

				CAPM		FF3M		FF5M		FF5M+MOM	
		Avg.Ret.	t-stat	α	t-stat	α	t-stat	α	t-stat	α	t-stat
		Panel A: PC1 of old-versus-new strategies ($\gamma'_{(t),1}(R_{X,(t-36),t+1} - \beta_{36} R_{X,(t),t+1})$)									
Unconditional	β^u_{36}	4.54	(4.00)	4.02	(3.57)	4.73	(4.64)	5.13	(4.70)	4.04	(3.62)
Conditional	$\beta^c_{36,t}$	4.92	(4.50)	4.46	(4.05)	4.85	(4.57)	4.63	(4.09)	3.76	(3.30)
		Panel B: Characteristics with returns that decay too fast ($R^{Fast}_{(t-36),t+1} = \gamma^{\leq 0,t}_{(t),1}(R_{X,(t-36),t+1} - \beta_{36} R_{X,(t),t+1})$)									
Unconditional	β^u_{36}	-3.69	(-3.79)	-3.30	(-3.40)	-4.06	(-4.55)	-4.37	(-4.57)	-3.55	(-3.57)
Conditional	$\beta^c_{36,t}$	-3.99	(-4.19)	-3.67	(-3.82)	-4.07	(-4.32)	-3.86	(-3.83)	-3.29	(-3.15)
		Panel C: Characteristics with returns that decay too slow ($R^{Slow}_{(t-36),t+1} = \gamma^{>0,t}_{(t),1}(R_{X,(t-36),t+1} - \beta_{36} R_{X,(t),t+1})$)									
Unconditional	β^u_{36}	3.18	(3.27)	2.71	(2.77)	2.52	(2.74)	2.84	(2.96)	1.86	(1.96)
Conditional	$\beta^c_{36,t}$	3.47	(3.64)	2.93	(3.08)	2.90	(3.14)	2.91	(2.90)	1.75	(1.84)

Table 5: old-versus-new sorts when returns decay fast versus slow

Table 6

Old and new stocks in benchmark factor models. This table presents average returns for old and new stock strategies as well as the intercept (with corresponding t-statistic in parentheses) from regressing these returns on benchmark factor models. As discussed in Section 5.3, the returns of new, $R^{New}_{X,(t),t+1}$, and old, $R^{Old}_{X,(t),t+1}$, stocks together make up the return of the newest sort, $R_{X,(t),t+1}$. As in the previous tables, we aggregate the new and old stock returns across the 56 characteristics using the PC1 loadings. For instance, the aggregated return of old stocks presented in the first row of Panel A is defined as $\gamma'_{(t),1} R^{Old}_{X,(t),t+1}$. In Panel B and C, we decompose these test asset returns into the part coming from the characteristics with returns that decay too fast versus too slow, which are those characteristics on which PC1 loads with a negative versus positive sign (see Eqs. (22) and (23)). Because the absolute sum of negative loadings is about twice as large as positive loadings, we rescale the latter such that the old-minus-new stock return in Panel C has the same volatility as the old-minus-new stock return in Panel B. The sample period runs from July 1972 through December 2019.

		CAPM		FF3M		FF5M		FF5M+MOM	
	Avg. ret.	t-stat	α	t-stat	α	t-stat	α	t-stat	α	t-stat
	Panel A: PC1 of old and new stock strategies (e.g., $\gamma'_{(t),1} R^{Old}_{X,(t),t+1}$)									
Old	-1.41	(-0.35)	-8.22	(-2.49)	-6.85	(-3.27)	-0.04	(-0.02)	2.59	(1.54)
New	-5.74	(-1.78)	-11.40	(-4.33)	-11.09	(-5.66)	-4.95	(-2.97)	-1.87	(-1.14)
Old-minus-new	4.33	(2.74)	3.18	(2.04)	4.25	(3.15)	4.91	(3.54)	4.46	(3.05)
	Panel B: Characteristics with returns that decay too fast (e.g., $R^{Old,Fast}_{(t),t+1} = \gamma^{\leq 0,t}_{(t),1} R^{Old}_{X,(t),t+1}$)									
Old	3.67	(1.17)	9.15	(3.57)	7.76	(4.40)	2.22	(1.58)	-0.09	(-0.06)
New	6.36	(2.61)	10.79	(5.49)	10.20	(6.47)	5.37	(3.95)	3.20	(2.29)
Old-minus-new	-2.69	(-1.96)	-1.64	(-1.22)	-2.44	(-2.07)	-3.15	(-2.62)	-3.29	(-2.68)
	Panel C: Characteristics with returns that decay too slow (e.g., $R^{Old,Slow}_{(t),t+1} = \gamma^{>0,t}_{(t),1} R^{Old}_{X,(t),t+1}$)									
Old	3.69	(2.37)	1.53	(1.11)	1.50	(2.05)	3.57	(5.01)	4.11	(5.51)
New	1.01	(0.68)	-0.99	(-0.77)	-1.46	(-1.75)	0.69	(0.90)	2.19	(3.29)
Old-minus-new	2.69	(3.60)	2.53	(3.35)	2.96	(4.08)	2.88	(3.84)	1.92	(2.35)

Table 6: old and new stocks in benchmark factor models

5.7 Table 7: long-term discount rates

Read:

- Table 7 converts the horizon dynamics into implied long-term discount rates.
- Under the null, high-minus-low discount-rate differences are modest.
- When persistent components are compensated, discount-rate differences become much larger.
- In the data, firms loading strongly on slow-decay characteristics have larger discount-rate differences than firms loading strongly on fast-decay characteristics.
- Individual characteristics such as book-to-market and size imply larger long-term discount-rate spreads than profitability and investment.

Economic message: Persistent characteristic compensation matters for valuation. It implies that characteristics can create long-lasting discount-rate differences across firms.

Long-term discount rates in the model and data. In this table, we compare implied discount rates r from our model to the data (see Section 7 for more detail). The r is found solving: $P = \frac{D}{r-g}$, where we set $D = 1$, $g = 1\%$, and the fundamental price P is calculated by discounting the implied cash flow stream using a hypothetical term structure of per-period discount rates derived from realized annual returns of a characteristic-sorted portfolio. We track realized annual returns up to 10 years after portfolio formation and assume that the per-period discount rate has converged back to the expected market return of 8% after year 10. In Panel A, we report the simulated distribution of implied discount rates for the high- and low-decile portfolio as well as the high-minus-low difference for the three cases in the model of Section 4. Under the null of a standard characteristic-based model of expected returns, the compensation for the transitory and persistent component of the characteristic is equal: $\lambda_P = \lambda_T = 3.17\%$. Under the first alternative, only the transitory component is compensated in expected returns: $\lambda_P = 0$, $\lambda_T = 4.49\%$. Under the second alternative, only the persistent component is compensated in expected returns: $\lambda_P = 4.49\%$, $\lambda_T = 0$. In Panel B, we report the implied discount rate from the data when we sort stocks on two composite characteristics that average at the firm level over the subset of characteristics on which PC1 loads with a positive versus negative sign (and the decay in returns is fast versus slow). In Panel C, we report the implied discount rate from the data when we sort stocks on a single characteristic (size, book-to-market, profitability, and investment). The sample period runs from July 1972 to December 2019. In Panels B and C, we report results for both value- and equal-weighted portfolios.

		High			Low			High-minus-low		
		Panel A: Model simulations								
		5th	50th	95th	5th	50th	95th	5th	50th	95th
Null:	$\lambda_P = \lambda_T > 0$	6.85%	8.86%	11.30%	5.65%	7.21%	9.08%	0.65%	1.57%	2.99%
Only transitory:	$\lambda_T > \lambda_P = 0$	6.28%	8.12%	10.38%	6.20%	7.85%	9.89%	-0.63%	0.26%	1.20%
Only persistent:	$\lambda_P > \lambda_T = 0$	7.02%	9.08%	11.56%	5.52%	7.03%	8.87%	1.05%	1.97%	3.35%
		Panel B: Composite characteristics (Data)								
		Value-weighted								
Fast		7.86%				8.13%				-0.27%
Slow		9.33%				6.83%				2.51%
		Equal-weighted								
Fast		7.88%				8.11%				-0.23%
Slow		9.93%				6.39%				3.53%
		Panel C: Individual characteristics (Data)								
		Value-weighted								
Size		8.79%				7.27%				1.52%
Book-to-market		9.04%				7.06%				1.97%
Profitability		8.17%				7.82%				0.35%
Investment		8.08%				7.92%				0.15%
		Equal-weighted								
Size		9.11%				7.00%				2.11%
Book-to-market		9.60%				6.63%				2.98%
Profitability		8.33%				7.68%				0.65%
Investment		8.90%				7.17%				1.73%

6 Short summary

- Firm characteristics contain both persistent and transitory components.
- The paper asks whether these components are compensated equally in expected returns.
- The null is $\lambda_P = \lambda_T > 0$.
- Old sorts are portfolios formed using characteristics measured 12 to 60 months earlier.
- If the null holds, old sorts should have zero alpha relative to newest sorts after controlling for remaining exposure.

- In the data, old-versus-new alphas vary strongly in sign and magnitude across characteristics.
- Positive alphas imply relatively high compensation for persistent components.
- Negative alphas imply relatively high compensation for transitory components.
- Old-minus-new stock strategies confirm the same pattern within current characteristic portfolios.
- Benchmark factor models do not explain these horizon dynamics.
- The results imply large and economically meaningful long-term discount-rate differences for characteristics with persistent compensation.

7 Conclusion

7.1 Core conclusion

Baba-Yara, Boons, and Tamoni show that the predictive content of firm characteristics depends on whether the characteristic signal is persistent or transitory. The same current characteristic value can imply different expected returns depending on its history.

7.2 Economic intuition

If a firm's characteristic value is persistent, it may represent a durable risk, valuation, or mispricing component. If the value is transitory, it may represent a short-lived signal. Because the paper finds that both components can be compensated, and that the relative compensation varies by characteristic, a single characteristic-sorted factor cannot capture all relevant horizon dynamics.

7.3 Takeaway

The paper pushes characteristic-based asset pricing beyond one-month post-formation returns. A successful asset pricing model must explain both the newest sort and older sorts, both persistent and transitory components, and both short-run returns and long-term discount rates.